

How can quantum algorithms provide insight into economic outcomes?

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Abstract: My fascination with economics centers on the desire to simulate and predict outcomes in real-life situations. This paper synthesizes and analyzes information on quantum algorithms in relation to game theory, Monte Carlo simulations, and algorithms that utilize entanglement to answer the question: How can quantum algorithms provide insight into economics? An algorithm is most typically defined as a set of instructions given to a computer in order to solve a particular problem, and economists use algorithms to analyze behavior, risk, and trends at the micro and macroscopic levels. This paper dives into game theory and how Quantum Monte Carlo simulations can model risk and uncertainty. This comes down to the unique capabilities of qubits, such as exponential computational power, entanglement, and superposition. These properties enable quantum algorithms to solve problems traditionally difficult for classical computers. This paper also includes a quantum Penny Flip game simulation I made, which serves as a visualization of quantum algorithms in action. Quantum algorithms are useful for economics when the outcome is mathematically complex and when it is difficult to find an analytical solution [7]. However, one must note that this frontier is fairly rudimentary and that many of these promised benefits are not yet practical as they require more qubits than we currently have [7]. All in all, though, research into quantum algorithms suggests value in the field of economics with a potentially quadratic speed up in convergence rate for Monte Carlo Simulations and novel strategies for game theory scenarios that produce new equilibria with higher payoffs [1,2,3,5,6,7].

PQ Penny Flip [1,5]: The field of game theory is integral to economics because it models decision-making when strategic outcomes depend on others' decisions. In the paper "Quantum Strategies," Meyer discusses quantum algorithms and their applications to game theory. To achieve this, Meyer devises a game called "PQ Penny Flip," which is a zero-sum, finite game with no deterministic solutions in which one player (Q) uses a quantum strategy while the other (P) uses a classical strategy. Q wins if the coin ends on heads, and P wins if the coin ends on tails after a series of blind coin flips. With this game, Meyer is able to prove that a quantum strategy will always beat a classical strategy via superposition. This quantum strategy uses the Hadamard gate to manipulate the probability of the penny being heads or tails.

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad F = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \text{and} \quad H = \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{bmatrix}$$

FIG 1 [5]: The operators I and F (also known as X) correlate to not flipping and flipping, respectively. H represents the Hadamard gate.

First, by changing the probability to 50% so that the classical players' next turn will always result in a 50% chance of the coin being heads or tails, then, on their last turn, applying the Hadamard gate again to change the probability back to being 100% heads, guaranteeing a victory in this classically fair game. To better illustrate this process, see the simulation of the penny flip game in Colab Notebooks (the repository can be found [here](#)). In game theory, each classical player should traditionally have a 50% chance of winning, but a quantum algorithm can guarantee a win in this simple zero-sum game. In the field of economics, economists need to represent the randomness observed in real life using

probabilities, and since quantum algorithms allow for a superposition of outcomes, they can create new predictions and strategies that classical computers simply can not replicate. These new quantum strategies change the Nash equilibrium and, therefore, how we model and interpret strategic interactions in traditional market systems.

CHSH and the Prisoner's Dilemma [2,3]: Moreover, in H. Wang's paper "Advantages and Applications of Quantum Gate Theory", he explains how quantum algorithms allow for more possible strategies in traditional game theory games. This is due to their unique properties of both entanglement and superposition. Entangled bits can affect the state of another qubit regardless of distance, also known as spooky action at a distance. The only caveat is that you can't control the information that is sent via entanglement, and you need traditional communication channels to control/interpret the information. This means that the information transfer between entangled bits can't really occur at the speed of light. The CHSH game, named after its authors John Clauser, Michael Horne, Abner Shimony, and Richard Holt, demonstrates that qubits can communicate more effectively than classical bits by 10% when entanglement is introduced. This is because entanglement allows the outcomes of two qubits to be interconnected. In another game, the prisoner's dilemma, quantum strategies effectively break the traditional Nash equilibrium (i.e., when every player stands to be worse off by changing their strategy) and raise the payoff from one to three. When entanglement between players in this game is maximized, the entanglement parameter $\gamma = \pi/2$. In this case, the Nash equilibrium ceases to be for each player to defect. The new Nash equilibrium is for both players

to employ a quantum strategy: $\mathbf{Q} \otimes \mathbf{Q}$, with a payoff of 3. **Possible figures to introduce to explain

what is going on more:

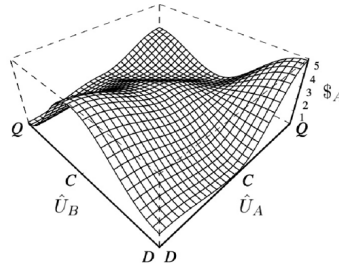
$$\hat{U}(\theta, \phi) = \begin{pmatrix} e^{i\phi} \cos \theta/2 & \sin \theta/2 \\ -\sin \theta/2 & e^{-i\phi} \cos \theta/2 \end{pmatrix} \quad \hat{Q} \equiv \hat{U}(0, \pi/2), \quad \hat{Q} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} **$$


FIG 2 [2]: Player A's payoff for a maximally entangled game

The overall payoff of both players under the quantum Nash equilibrium is higher than that of the classical one. Using quantum methods, players are better off. This relates to economics in that economists can implement quantum algorithms to create and predict not only new outcomes, but better outcomes. These new strategies bring insight into how oligopolies and international agreements work and how they could potentially improve outcomes. Research surfaces the theme that quantum algorithms bring novel approaches to solving problems related to economics.

Monte Carlo Simulations [4,6,7]: Shifting back to more concrete applications of quantum algorithms in the field of economics, in the paper “Quantum Monte Carlo for economics,” scientists analyze the benefits and drawbacks of quantum Monte Carlo simulations. Monte Carlo simulations use a cumulation of random sampling to model uncertainty, forecast economic trends, and assess risk. Quantum Monte Carlo, also known as QMC, has an error convergence rate of $O(N^{-1})$, which is much faster than that of a classical Monte Carlo simulation, $O(N^{-1/2})$. Likewise, the number of samples (N) required for QMC estimation is proportional to one over the target error, $\frac{1}{\epsilon}$ while for

MC estimation, N is proportional to $\frac{1}{\epsilon^2}$, meaning that QMC can estimate an expected value using fewer samples than a classical MC. This is all because QMCs can apply the quantum amplitude estimation algorithm to obtain this speedup. This indicates that quantum algorithms can estimate economic outcomes more accurate and faster than classical ones.

Conclusion: Quantum computers struggle to juggle high fidelity, fast gate times, coherence time, and scalability all at once, which in turn limits quantum algorithm benefits and applications. While quantum algorithms have promise in the world of economics, they are difficult to implement and, therefore, generally less practical for solving economic problems than classical computers as of now. Overall, quantum algorithms, despite their limitations, provide new insight with the potential to alter the landscape of financial modeling and strategies.

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